

Aviral Asthana

aviral.asthana070402@gmail.com | +91-8376976203 | linkedin.com/in/aviral-asthana-02b70824b | GitHub | Portfolio

Technical Skills

Programming Languages: Go, Java, Python, SQL, C++

AI-Native Tooling: Cursor, Claude (daily use for shipping production code)

Backend & Frontend: Spring Boot, Node.js, React.js, Next.js

Event-Driven & Data: Kafka, AWS SQS, AWS S3, PostgreSQL, MySQL

Fintech/Banking: Core Banking System (CBS) Integration, KYC/Onboarding, RBAC, Workflow Orchestration

Cloud & DevOps: AWS (ECS, CloudWatch, Secrets Manager), Docker, Kubernetes, Git, CI/CD, Linux

Experience

Backend Software Engineer, Aspura

April 2025-Present

- Independently built the end-to-end **RFI (Request for Information)** flow **from 0 to 1** – solo-designed a backend-driven **workflow orchestrator** powering both web and **mobile** screens, shipped it, and mentored another engineer along the way. Now live for **500+** users.
- Owned **Go**-based backend services for Aspura's **NRI Banking** product end to end, integrating directly with the **Core Banking System (CBS)** for onboarding, KYC, and transactions – took the product from zero to **1,000+** real bank accounts opened.
- Built an in-house **AI-powered OCR system** from scratch: used the **Adapter pattern** + **feature flags** to orchestrate across **LlamaParse v2**, **Claude Sonnet**, and **Landing AI**, killing a recurring third-party vendor bill (Persona) and letting the product route each document to whichever model is cheapest or best.
- Contributed to two event-driven services on **Kafka/AWS SQS**: a multi-tenant **Communications Service** (multi-vendor email/SMS, domain classification, retries, concurrent multi-channel delivery) and a compliance-grade **Audit Service** (event bus → **S3** → REST trails by user/event).
- Designed and shipped **Pulse**, an internal CRM in **Java/Spring Boot** with bank-grade **RBAC** secure enough for external banking partners to use directly – **50+** agents, sub-**1-day** ticket TAT.
- Use **Cursor** and **Claude** daily to move faster end to end – scaffolding services, writing tests, and shipping the above features solo without waiting on a bigger team.
- Took a critical API endpoint from **10+ seconds to <250ms** (40x) by hand – **CompletableFuture** for parallel processing, smarter indexes, and a **Redis** caching layer – unblocking a manual release process that was quietly costing the team hours every week.
- Shipped multiple **React** dashboards on the **Alphadesk** platform from scratch for real-time order and transaction monitoring, so ops teams could see problems as they happened instead of after the fact.

Web Developer Intern, Codestam Technologies

March 2024-April 2024

- Developed backend services and **RESTful APIs** using **Node.js** and **Express.js**, working as part of a team to deliver projects **20%** faster through efficient backend architecture.
- Built real-time backend services for chat applications using **WebSockets**, developed e-commerce APIs with payment processing, and created administrative management system backends with **PostgreSQL** and **MongoDB**, serving **1,000+** active users.

Projects

QueryWise – Cloud DB Cost Optimization Platform

QueryWise GitHub

- Built **QueryWise**, a single-binary **Go** CLI that reads PostgreSQL's `pg_stat_statements` and ranks the most expensive query patterns by a configurable, weighted cost heuristic across execution time, I/O, and call frequency.
- Designed a **privacy-first architecture**: query text is fingerprinted locally via **SHA-256/HMAC** the instant it's read and discarded, so raw SQL is never stored or transmitted; enforced **TLS by default** and required only the narrow `pg_read_all_stats` grant instead of superuser access.
- Integrated optional **Claude API**-powered tuning recommendations that send only query fingerprints and numeric metrics (never SQL), and added **terminal**, **Markdown**, and **JSON** report exports for CI and team sharing.
- Built this entirely on my own time because existing tools forced a tradeoff between visibility and privacy that didn't need to exist – open-sourced it on GitHub for anyone running Postgres in production to use.

Education

Indian Institute of Information Technology, Nagpur, B.Tech in Computer Science

Nov 2022 - May 2026

- CGPA: 7.8/10.0